

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    roc_auc_score,
    confusion_matrix,
    classification_report,
    roc_curve
)

# Load Dataset
df = pd.read_csv("data.csv")

# Remove unnecessary columns
df.drop("id", axis=1, inplace=True)
df.drop("Unnamed: 32", axis=1, inplace=True)

# Convert diagnosis column
le = LabelEncoder()
df["diagnosis"] = le.fit_transform(df["diagnosis"])

# Features and Target
X = df.drop("diagnosis", axis=1)
y = df["diagnosis"]

# Train Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Standardization
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Logistic Regression Model
model = LogisticRegression(max_iter=1000)

model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Probabilities
y_prob = model.predict_proba(X_test)[:,1]

# Evaluation Metrics
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
roc_auc = roc_auc_score(y_test, y_prob)

print("Accuracy :", accuracy)
print("Precision :", precision)
print("Recall :", recall)
print("ROC-AUC :", roc_auc)

# Confusion Matrix
```

```
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix")
print(cm)

# Classification Report
print("\nClassification Report")
print(classification_report(y_test, y_pred))

# Heatmap
plt.figure(figsize=(6,4))

sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues'
)

plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.show()

# ROC Curve
fpr, tpr, thresholds = roc_curve(y_test, y_prob)

plt.figure(figsize=(6,4))

plt.plot(
    fpr,
    tpr,
    label=f"AUC = {roc_auc:.3f}"
)

plt.plot([0,1],[0,1], 'r--')

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")

plt.legend()

plt.show()

print("Task 4 Completed Successfully")
```

Accuracy : 0.9736842105263158
Precision : 0.9761904761904762
Recall : 0.9534883720930233
ROC-AUC : 0.99737962659679

Confusion Matrix

```
[[70  1]
 [ 2 41]]
```

Classification Report

	precision	recall	f1-score	support
0	0.97	0.99	0.98	71
1	0.98	0.95	0.96	43
accuracy			0.97	114
macro avg	0.97	0.97	0.97	114
weighted avg	0.97	0.97	0.97	114

